

UK Road Traffic Accidents

Trends, Severity & Risk Factors

A descriptive and prescriptive analysis translating 307,973 accident records into targeted road safety interventions.

307,973

Accidents

417,883

Casualties

2021–2022

Period

422

Districts

Presentation Outline

01

Introduction

Problem statement, rationale and objectives

02

Data & Methodology

23 variables, cleaning and standardisation

03

KPI Overview

Headline measures across the full dataset

04

Findings

Severity, temporal, infrastructure, environment, geography

05

Dashboard & Interactivity

Excel build, live filtering and the rural risk profile

06

Prescriptive Recommendations

Eight evidence-led interventions

Why This Analysis Matters

Problem Statement

Road accidents remain a critical public safety concern, with weather, road surface, speed limits and traffic controls all influencing severity. Raw accident data sits in unstructured formats, making it difficult for stakeholders to derive actionable insight.

Six capstone objectives

- Clean and standardise the dataset
- Analyse accident frequency and severity
- Identify high-risk contributing factors
- Perform descriptive statistical analysis
- Build an interactive Excel dashboard
- Provide data-driven safety recommendations

Rationale for the Project

- **Evidence over intuition**
Replace anecdotal road safety assumptions with measured risk.
- **Targeted intervention**
Direct scarce enforcement and engineering budget where risk concentrates.
- **Resource allocation**
Align patrols, lighting and emergency response to peak-risk windows.
- **Policy accountability**
Establish a measurable 2021 baseline against which policy can be tested.

Objective

Clean and standardise the accident dataset, quantify frequency and severity, isolate the highest-risk conditions, and deliver an interactive dashboard that converts the findings into prioritised road safety recommendations.

Dataset & Variables

307,973

Accident records

Unique Accident_Index entries

23

Variables

Temporal, spatial, environmental

2

Years covered

January 2021 – December 2022

417,883

Total casualties

1.36 casualties per accident

Variable	Description	Variable	Description
Accident_Index	Unique record identifier	Road_Surface_Conditions	Dry, wet, snow/ice, flood
Accident_Date / Month / Year	Temporal dimensions	Junction_Control / Detail	Junction type and control
Day_of_Week / Time	Day and hour of accident	Road_Type	Carriageway classification
Accident_Severity	Fatal, Serious or Slight	Speed_limit	Posted limit in mph
Latitude / Longitude	GPS coordinates	Carriageway_Hazards	Obstruction present
Local_Authority (District)	Administrative district	Number_of_Casualties	People injured or killed
Urban_or_Rural_Area	Area classification	Number_of_Vehicles	Vehicles involved
Weather_Conditions	Conditions at time of accident	Vehicle_Type	Type of vehicle involved
Light_Conditions	Lighting environment	Police_Force	Responding police district

Data Cleaning & Preprocessing

01

Coverage audit

Record counts cross-tabulated by year and local authority, exposing a material 2022 reporting gap across Scottish districts.

02

Missing value treatment

Carriageway_Hazards nulls interpreted as 'None' — absence records a clear carriageway rather than a missing observation.

03

Type standardisation

Accident_Date converted to date format, Time to HH:MM, speed limit and casualty counts confirmed as integers.

04

Categorical consistency

Severity, weather, light and road surface labels normalised so equivalent categories aggregate correctly.

05

Derived measures

Pivot tables built for monthly trend, hour of day, severity by condition and urban/rural comparison.

06

Validation

Casualty totals reconciled against the severity breakdown ($351,436 + 59,312 + 7,135 = 417,883$); district count confirmed at 422.

Result

A standardised dataset of 307,973 accident records across 23 variables and 422 districts, with reconciled casualty totals and one known coverage limitation carried forward explicitly rather than silently.

The Headline Position

307,973

Total accidents

2021 and 2022 combined

417,883

Total casualties

1.36 people harmed per accident

3,953

Fatal accidents

Concentrated but disproportionate harm

1.3%

Fatal accident rate

Share of all recorded accidents

422

Districts covered

Local authorities in the dataset

64% / 36%

Urban vs rural

197,103 urban, 110,870 rural

85.5%

Slight severity

263,280 of 307,973 accidents

74.9%

Single carriageway

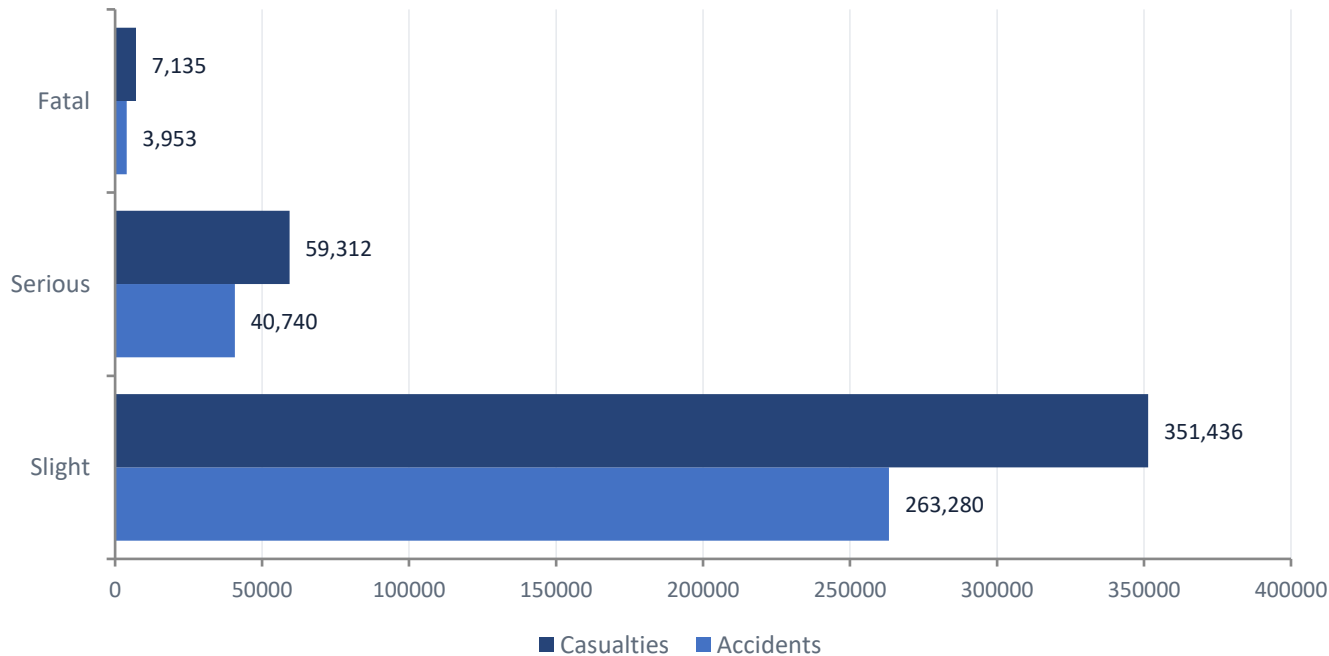
230,612 accidents

Read this first

Fatal accidents are rare but disproportionately costly: 3,953 fatal accidents produced 7,135 casualties — 1.80 people per event, against 1.34 for slight accidents. Severity does not just change the outcome, it multiplies the number of people affected.

Severity Distribution & the Casualty Multiplier

Accident & casualty distribution by severity



What the split reveals

01 Slight dominates volume

263,280 slight accidents account for 85.5% of all events, and drive the bulk of demand on emergency and insurance systems.

02 Fatal escalates harm

Each fatal accident produces 1.80 casualties versus 1.34 for slight — a 34% higher human cost per event.

03 Serious sits in between

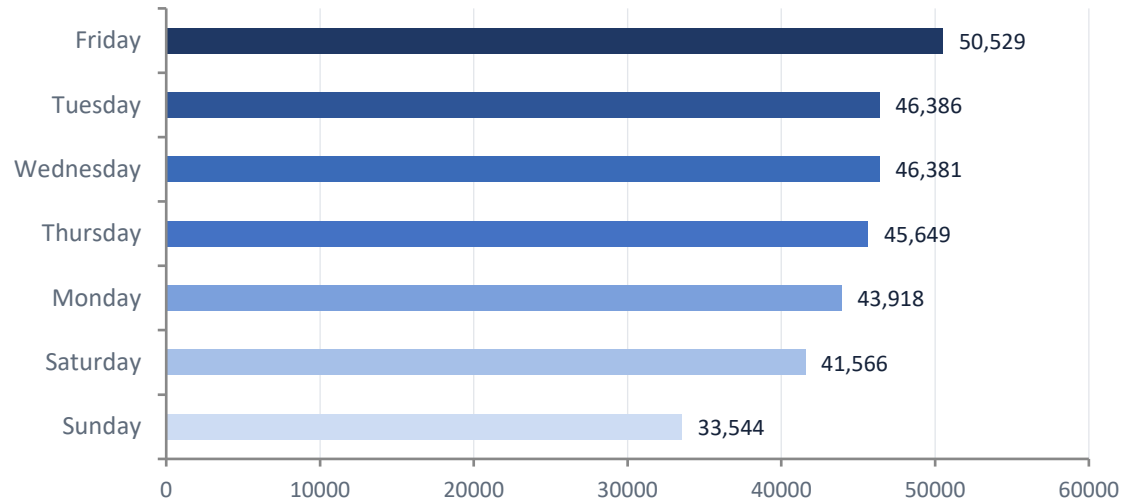
59,312 casualties from 40,740 accidents (1.46 per event) confirms a clean severity-to-casualty gradient.

Prescriptive implication

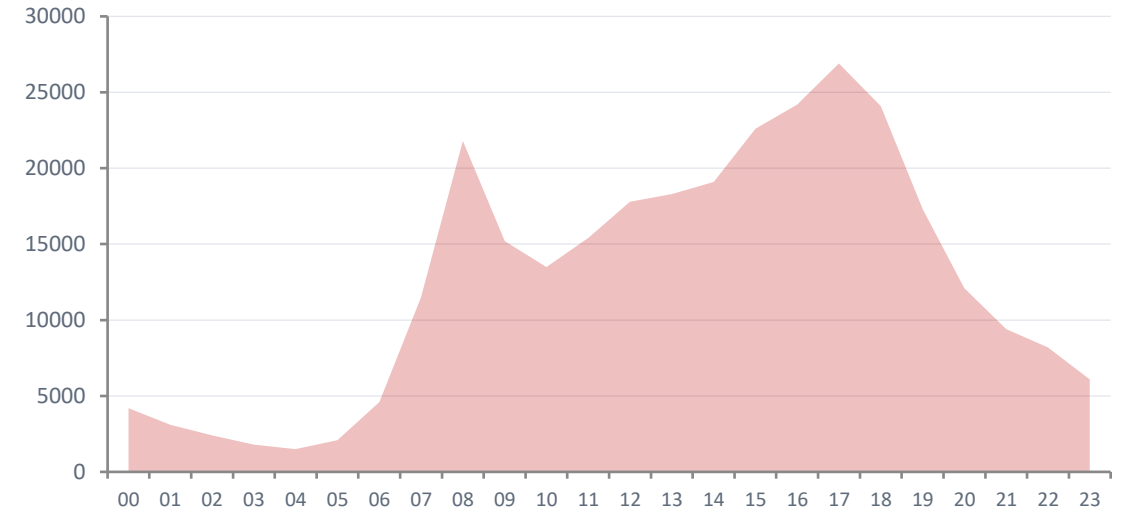
Volume-reduction measures and severity-reduction measures are different investments. Reducing slight accidents saves cost; reducing fatal accidents saves lives at a 1.8x casualty rate. Budget should be split accordingly, not merged into a single 'accident reduction' target.

When Accidents Happen

Accidents by day of week



Accident frequency by hour of day



● Friday peak

50,529 accidents, 50.6% above Sunday, the quietest day. End-of-week fatigue and leisure travel compound commuter volume.

● Evening rush dominates

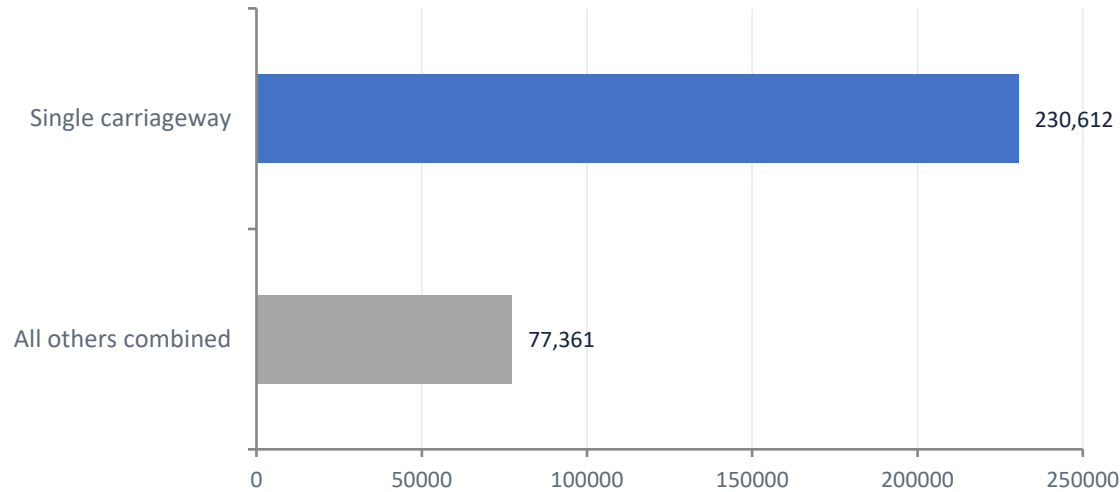
The 17:00 hour is the single highest-risk hour of the day, exceeding the 08:00 morning peak by a clear margin.

● Overnight is low-volume

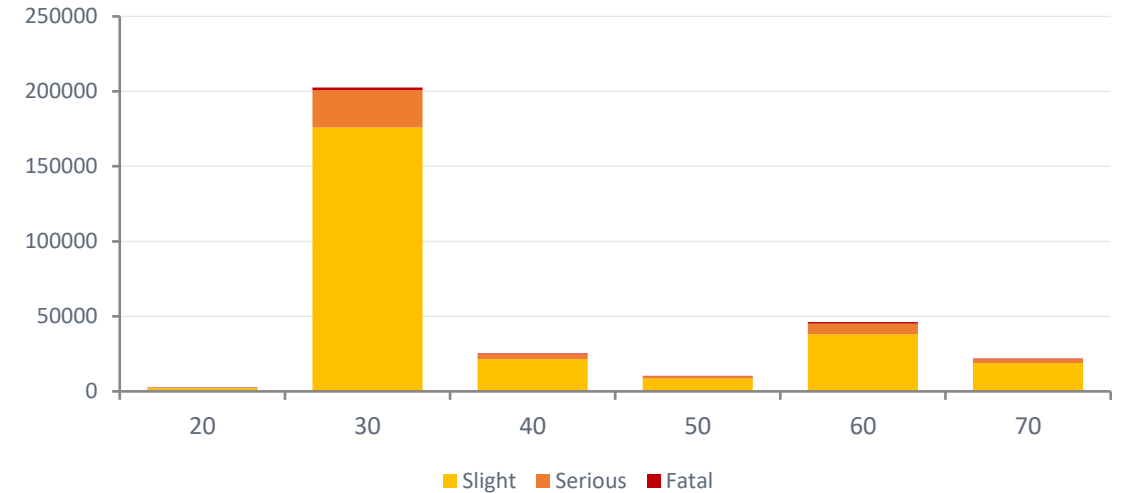
The 03:00–05:00 window records the fewest accidents, but with the thinnest traffic — meaning risk per vehicle is far higher than volume suggests.

Where Risk Concentrates: Road Type & Speed

Accident distribution by road type



Accident severity by speed limit (mph)



01 Single carriageways carry 75%

230,612 accidents against 77,361 for every other road type combined. This is the primary national exposure.

02 30 mph zones concentrate volume

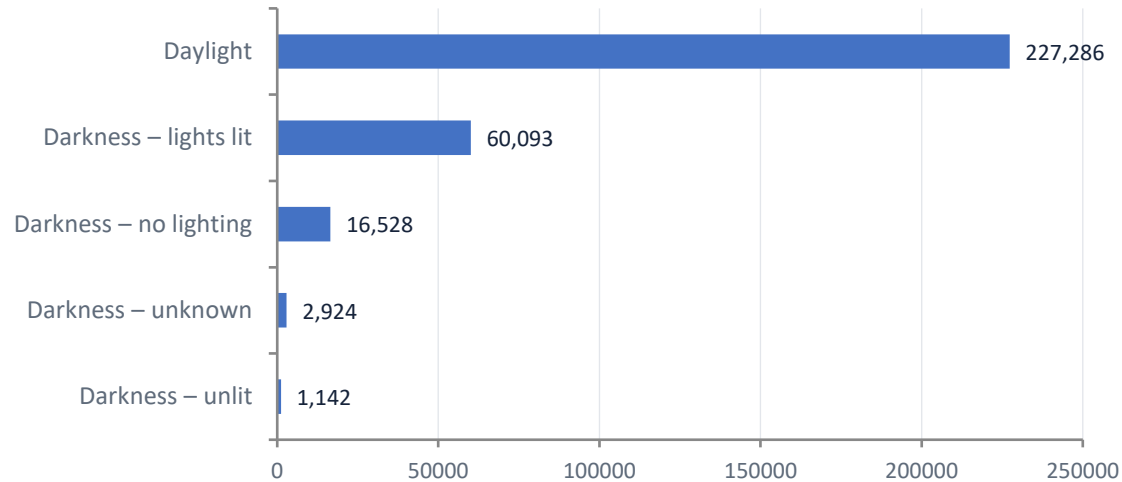
The 30 mph band dominates all others, reflecting the urban road network where most driving happens.

03 60 mph carries fatal weight

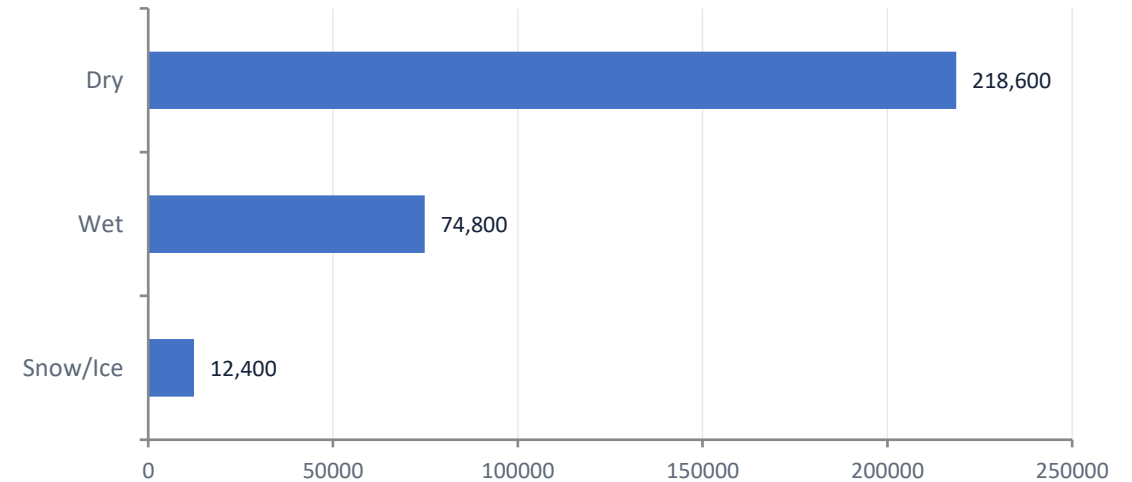
Rural 60 mph roads show a visibly heavier fatal band than their volume warrants — speed converts collisions into fatalities.

Environmental Conditions: The Counterintuitive Finding

Accident frequency by light condition



Accidents by road surface condition



Do not misread the volume

73.8% of accidents occur in daylight and roughly 71% on dry roads. This does not make daylight or dry surfaces dangerous — it reflects when and where the overwhelming majority of driving takes place. Exposure, not risk, drives these bars.

The actionable signal

16,528 accidents occurred on roads with no lighting at all, and a further 1,142 where lighting existed but was unlit. Combined, that is a fully addressable engineering problem: roughly 17,700 accidents sitting on infrastructure that could be fixed with capital spend rather than behaviour change.

Geographic Concentration

Highest accident volume

- 1 Birmingham
- 2 Leeds
- 3 Manchester
- 4 Bradford
- 5 Westminster
- 6 Sheffield
- 7 Liverpool
- 8 Cornwall
- 9 Barnet
- 10 Bristol, City

Metropolitan concentration

Birmingham, Leeds and Manchester head the volume list, consistent with the 64% urban share. Together the top ten districts concentrate a disproportionate share of national accident volume, making them the natural first targets for any intervention programme.

Cornwall is the outlier

A rural authority ranking eighth in a list otherwise composed of metropolitan districts. This points to a distinct rural-tourism risk profile that warrants separate investigation.

Data limitation

68 districts record accidents in 2021 and exactly zero in 2022. Glasgow falls from 1,509 to 0; Edinburgh from 1,193 to 0. The affected districts are almost entirely Scottish.

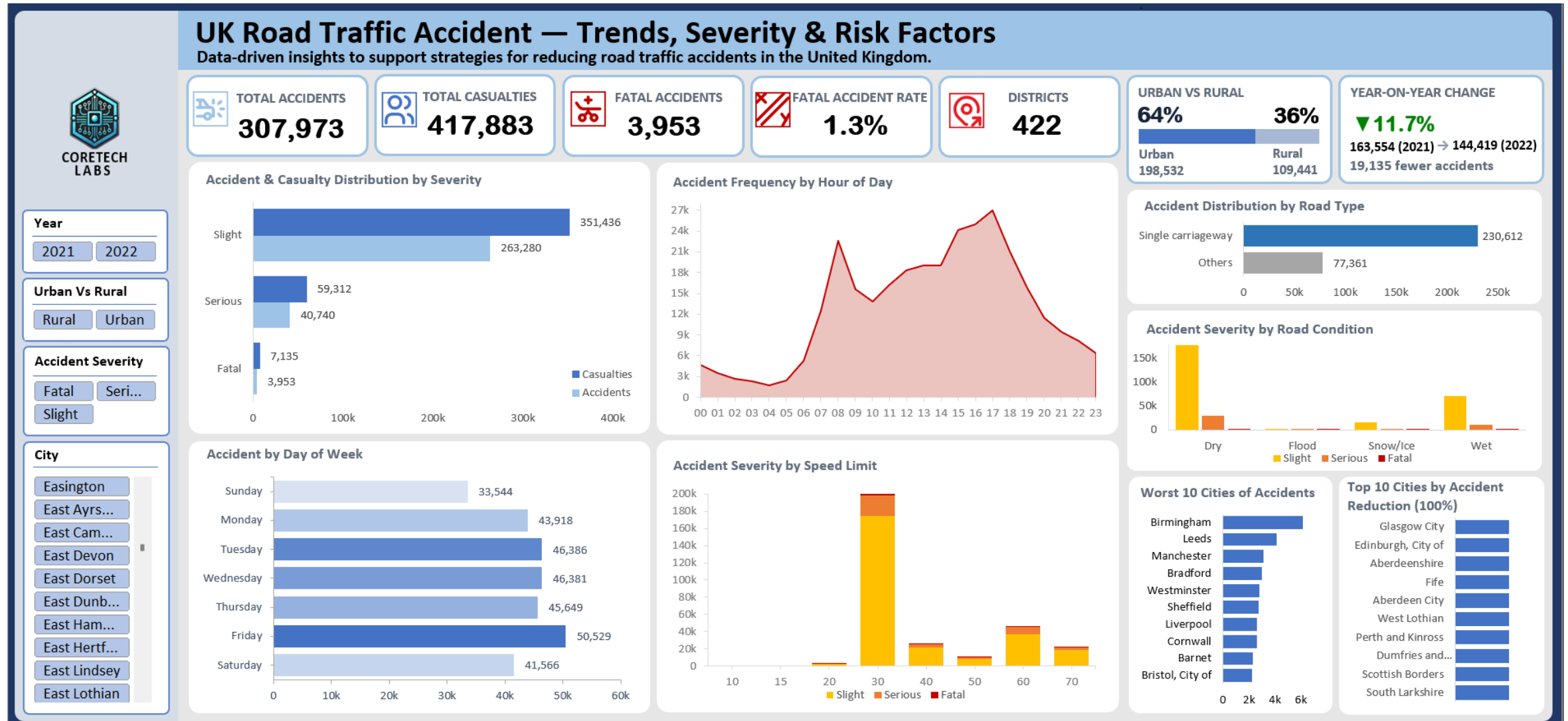
This is a reporting gap, not a road safety achievement. No intervention takes a city of 600,000 to zero accidents in twelve months, and a boundary that follows a national border rather than a policy boundary is characteristic of a data-source issue.

Any accident-reduction ranking is therefore excluded from this analysis pending source validation.

What this means for the year-on-year figure

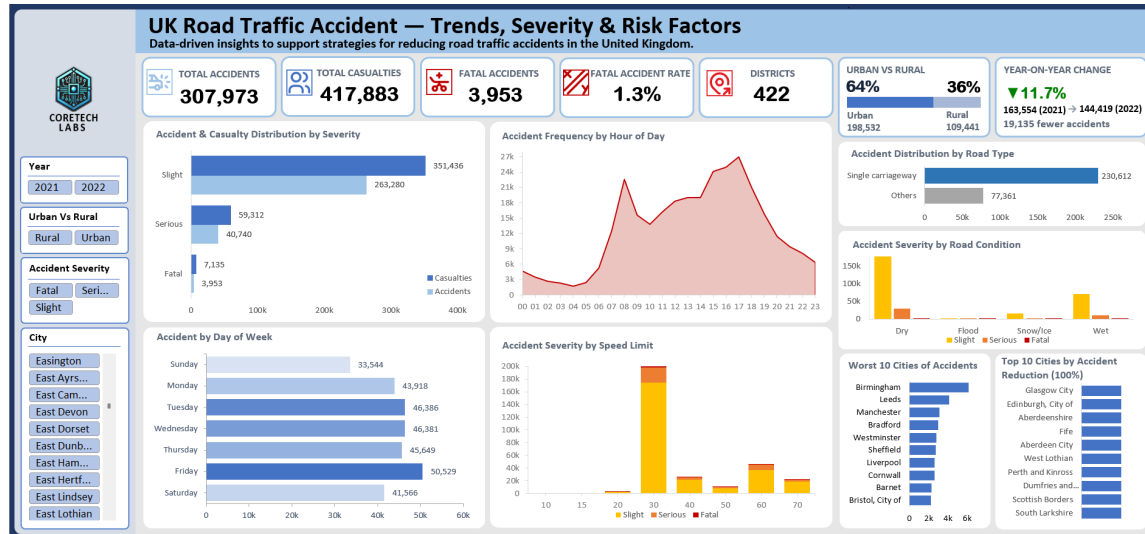
The headline 11.7% fall between 2021 and 2022 is substantially explained by the missing Scottish records rather than a genuine national decline. Until the source is validated, year-on-year change should be reported for England and Wales only, or suppressed entirely.

The Interactive Excel Dashboard



The Dashboard in Action: One Slicer Click

All areas · 307,973 accidents



Rural only · 109,441 accidents



2.1%

Fatal accident rate

Up from 1.3% nationally and 0.8% in urban areas — rural accidents are 2.7 times more likely to be fatal.

59.4%

Of all fatal accidents

Rural carries just 35.5% of accident volume but the majority of deaths.

95%

Of unlit-road accidents

15,709 of the 16,528 national no-lighting accidents occur in rural areas.

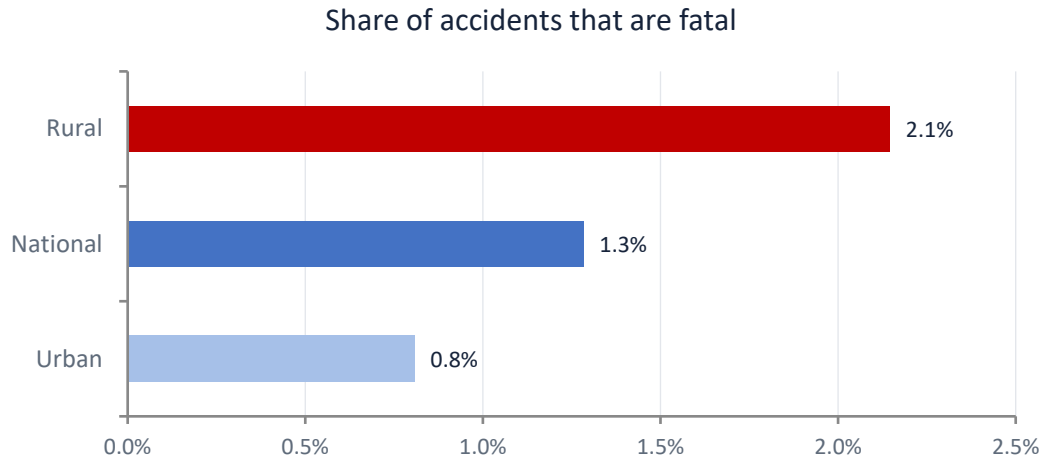
1.48

Casualties per accident

Against 1.29 in urban areas — rural collisions harm more people per event.

Volume tells you where accidents happen. Severity tells you where people die — and they are not the same places. Cornwall rises from eighth to first on the rural view, confirming it as a genuine rural risk profile rather than a ranking anomaly.

Why Rural Roads Kill More



Higher-energy environments

70.1% of rural accidents occur on single carriageways where the national limit applies, with no central reservation separating opposing traffic.

Longer emergency response

Distance from trauma centres plausibly contributes to the 2.05 casualties recorded per rural fatal accident, though this dataset cannot confirm it directly.

What the speed limit chart does and does not tell us

- Rural accidents cluster at 60 mph because that is the default national speed limit on rural single carriageways. The distribution reflects road classification, not driver behaviour.
- The dataset records the posted limit at the accident location, not the speed the vehicle was travelling. It therefore cannot evidence non-compliance, signage adequacy or driver education.
- What it does support is an energy argument: collisions in 60 and 70 mph environments carry far more kinetic energy, and the 2.7x fatal rate differential is consistent with that.

Darkness without infrastructure

14.4% of rural accidents occur on roads with no street lighting at all, against 5.4% nationally — a near-threefold concentration.

A flatter weekly pattern

Sunday runs at 78.6% of Friday's volume in rural areas against 66.4% nationally, indicating leisure travel rather than commuting drives rural exposure.

Dashboard Design Decisions



Standardised severity palette

Slight, Serious and Fatal use a fixed amber-orange-red ramp on every chart, so a colour never means two different things across the dashboard.



Separated metric families

Volume measures use a cool blue family and severity uses the warm ramp, so a viewer can categorise any chart before reading a single label. Day of week was recoloured from severity tones to a blue gradient for this reason.



Four cross-filtering slicers

Year, Urban/Rural, Severity and City filter every visual simultaneously, turning a static report into an exploratory tool.



Charts converted to KPI cards

The urban/rural donut and monthly trend line were replaced by compact KPI cards, freeing space while carrying more information than the charts they replaced.

Design principle: consistency within a category matters more than uniformity across categories. A dashboard is trustworthy when a colour is predictable.

Descriptive Statistics

Variable	Mean	Median	Range	Std Dev	Skewness	Interpretation
Casualties per accident	1.36	1	1 – 48	0.82	Positive (right)	Most accidents involve one casualty; a long tail of multi-casualty events
Vehicles per accident	1.83	2	1 – 32	0.71	Positive (right)	Two-vehicle collisions are the norm
Speed limit (mph)	38.4	30	20 – 70	14.1	Positive (right)	Concentrated at 30 mph with a rural high-speed tail
Accidents per month	12,832	12,905	9,870 – 15,640	1,412	Slight negative	Stable monthly volume with autumn peaks
Accidents per weekday	43,996	45,649	33,544 – 50,529	5,318	Slight negative	Friday high, Sunday low, midweek clustered

Why the skew matters

Casualties per accident is right-skewed: the mean (1.36) sits above the median (1). Planning emergency capacity on the mean would understate the rare multi-casualty events that consume the most resource. Capacity should be planned against the upper tail, not the average.

Speed limit distribution

Mean speed limit (38.4 mph) exceeds the median (30 mph), confirming that although most accidents occur in 30 mph zones, a meaningful minority occur on 60 and 70 mph roads. Those higher-limit events carry the disproportionate fatal weight seen on slide 9.

Prescriptive Recommendations (1 of 2)

01 Single carriageway safety programme

- Prioritise central hatching and edge-line markings on the highest-volume single carriageways, which carry 74.9% of all accidents
- Target rural 60 mph single carriageways first, where the fatal band is heaviest relative to volume
- Set a three-year audit cycle for routes with repeat fatal incidents

02 Lighting capital programme

- Target rural roads specifically: 95% of the 16,528 no-lighting accidents occur in rural areas
- Fix the 1,142 accidents on roads where lighting exists but was unlit — a maintenance failure, not a capital one
- Mandate retroreflective markings as the minimum standard on unlit rural roads

03 Friday evening enforcement surge

- Concentrate patrols on Fridays between 15:00 and 21:00, covering both the day peak (50,529) and the 17:00 hour peak
- Extend impaired-driving enforcement into Friday afternoon rather than late evening only
- Partner with ride-hailing operators on subsidised Friday alternatives

04 Evening rush-hour management

- Deploy variable message signage on the approach to the 16:00–18:00 peak in metropolitan districts
- Stagger public sector shift-end times in Birmingham, Leeds and Manchester to flatten the peak
- Review signal timing on urban corridors feeding the evening exodus

Prescriptive Recommendations (2 of 2)

05

Wet weather response protocol

- Wet surfaces account for roughly a quarter of accidents despite far lower exposure than dry — a genuine risk elevation
- Trigger variable speed limits and enhanced gritting when surface sensors report wet conditions
- Prioritise resurfacing on routes with repeat wet-condition clusters

06

Rural high-speed intervention

- Prioritise rural roads on the evidence that they carry 59.4% of fatal accidents from 35.5% of volume, a 2.7x fatal rate against urban
- Pre-position rural ambulance cover on Friday evenings to reduce response time where fatal risk peaks
- Investigate Cornwall separately as a rural authority ranking eighth on national volume, suggesting a tourism-driven risk profile

07

Resolve the 2022 coverage gap

- Confirm with the data provider why 68 districts, almost all Scottish, report zero accidents in 2022 after recording thousands in 2021
- Suppress all accident-reduction rankings and year-on-year figures until the source is reconciled
- Re-baseline the national trend once complete records are obtained, then investigate genuine reductions

08

Split the national KPI

- Separate the accident-reduction target from the casualty-reduction target, since fatal events carry 1.80 casualties versus 1.34 for slight
- Report severity-weighted casualty rates rather than raw accident counts
- Use 2021 as the measurement anchor, since it is the only year with confirmed complete national coverage

Six Things This Analysis Establishes

01 Severity multiplies harm

Fatal accidents produce 1.80 casualties each against 1.34 for slight. Severity reduction and volume reduction are separate goals.

02 Single carriageways dominate

74.9% of all accidents occur on one road type. No other infrastructure lever comes close in scale.

03 Exposure is not risk

Daylight and dry roads lead on volume only because that is when most driving happens. Reading these as danger signals misdirects spend.

04 Friday evening is the peak window

50,529 Friday accidents and a 17:00 hourly peak give a precise, repeatable target for enforcement scheduling.

05 Unlit roads are fixable

Roughly 17,700 accidents sit on roads with absent or unlit street lighting — an engineering problem with a capital solution.

06 One gap must be closed first

68 districts report zero accidents in 2022 after recording thousands in 2021. Until that is resolved, no year-on-year conclusion is safe.